

Unit 1 Course Pack

Basics of Quantum Technology

Quantum Entanglement & Quantum Computing (UE23EC343AB3)
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What Is In This Pack

Unit 1 of the syllabus, all fourteen hours, in twelve sections. **One section per syllabus topic, in syllabus order**, so you can read this alongside the course outline without translating.

Every section is built the same way:

1. a **blue box** giving the idea in plain words before any symbols;
2. the minimum mathematics needed to compute with;
3. **worked examples** in the same question-and-solution boxes as your question bank, so practice here looks exactly like the exam;
4. a **cyan problem box** with five problems.

Companion notebook: QEQC_Unit01_Lab.ipynb (Section 12).

The Whole Unit In Five Lines

If you take away only these five things, the rest of the course is followable.

1. Quantum mechanics adds *amplitudes* (complex numbers) and squares at the end. Classical physics adds probabilities directly. Everything else follows from that one change.
2. A state is a vector $|\psi\rangle$. Outcome i has probability $|c_i|^2$.
3. Gates and time evolution are **unitary**. Observables are **Hermitian**.
4. A qubit lives in $\mathbb{C}^2$ and draws as a point on a sphere.
5. Measurement destroys the superposition. One bit out, and the state collapses.

Prerequisites

Complex numbers in polar form, matrix multiplication, eigenvalues and eigenvectors, conjugate transpose, orthonormal bases. Nothing beyond that. If any of these is shaky, fix it before Section 4.

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1 Quantum and Classical Physics

The Idea

People usually say “classical physics is certain, quantum physics is random.” That is not the real difference. Classical physics uses probability constantly, in weather forecasting and in statistical mechanics.

The real difference is **how the randomness is bookkept**. Classical physics tracks probabilities, which are non-negative. Quantum physics tracks **amplitudes**, which are complex, and squares them only at the very end.

Because amplitudes are complex, two contributions can **cancel**. Two probabilities never can. That is the entire difference, and every quantum effect in this course comes out of it.

1.1 Two ways of counting

Suppose there are two ways for something to happen.

Classical	Quantum
Each route carries a probability $P_1, P_2 \geq 0$	Each route carries an amplitude $A_1, A_2 \in \mathbb{C}$
Total = $P_1 + P_2$	Total = $ A_1 + A_2 ^2$
Adding a route always helps	Adding a route can <i>cancel</i> the first

Classical state vs. quantum state

A classical system with n configurations is described by a probability vector

$$\vec{p} = (p_1, \dots, p_n), \quad p_i \geq 0, \quad \sum_i p_i = 1.$$

The same system quantum mechanically is described by an amplitude vector

$$\vec{c} = (c_1, \dots, c_n), \quad c_i \in \mathbb{C}, \quad \sum_i |c_i|^2 = 1,$$

and probabilities appear only on measurement, as $p_i = |c_i|^2$.

The One Line That Matters

$$|A_1 + A_2|^2 = |A_1|^2 + |A_2|^2 + \underbrace{2 \operatorname{Re}(A_1^* A_2)}_{\text{interference term}}$$

Classical physics is missing the last term. It can be positive (constructive) or negative (destructive), and it averages to zero, which is why interference is invisible unless you look carefully.

1.2 Four things that are genuinely new

1. **Superposition.** A system can occupy a combination of options. This is a real, definite state, not “we don’t know which.”
2. **Measurement disturbs.** Reading a quantum state changes it. There is no peeking.
3. **Incompatible quantities.** Position and momentum, or X and Z for a qubit, cannot both be sharp. Not an instrument limitation.
4. **Entanglement.** Two systems can have a definite joint state while neither has a definite state of its own. That is Unit 3.

A Tempting Wrong Answer

“The electron really has a definite position, we just don’t know it.” This sounds reasonable and is wrong. Bell tests (Unit 3) rule it out for any *local* theory. Treat superposition as a real physical state, not as missing information.

1.3 Worked examples

Example 1.1. Two routes each have amplitude of magnitude $1/\sqrt{2}$. Compute the total probability when (a) the amplitudes have the same sign, (b) opposite signs. Compare with the classical answer in both cases.

Solution.

(a) Same sign:

$$\left| \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right|^2 = |\sqrt{2}|^2 = 2.$$

(b) Opposite sign:

$$\left| \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right|^2 = 0.$$

Classically both cases give $\frac{1}{2} + \frac{1}{2} = 1$, with no way to reach either 2 or 0.

Why this matters. Same two routes, same individual strengths. The answer moved from maximum to zero purely because of a sign, i.e. a relative phase.

Example 1.2. A superconducting qubit has a $|0\rangle \rightarrow |1\rangle$ gap of 5 GHz. Estimate the probability of thermal excitation at 300 K and at 10 mK, and explain why quantum computers need a dilution refrigerator.

Solution.

Convert the gap to a temperature scale:

$$\frac{h \times 5 \times 10^9}{k_B} = \frac{6.63 \times 10^{-34} \times 5 \times 10^9}{1.38 \times 10^{-23}} \approx 0.24 \text{ K}.$$

The excitation probability goes as the Boltzmann factor $e^{-\Delta E/k_B T}$.

At $T = 300$ K:

$$e^{-0.24/300} = e^{-0.0008} \approx 0.999.$$

The qubit is a fair coin before you touch it: unusable.

At $T = 10$ mK:

$$e^{-0.24/0.01} = e^{-24} \approx 4 \times 10^{-11}.$$

Initialisation into $|0\rangle$ now succeeds essentially every time.

Conclusion. The fridge exists to make $k_B T \ll \Delta E$, so the qubit reliably starts in its ground state.

Example 1.3. A cricket ball of mass 160 g travels at 30 m/s. Find its de Broglie wavelength and comment on why it never diffracts.

Solution.

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{0.160 \times 30} = \frac{6.63 \times 10^{-34}}{4.8} \approx 1.4 \times 10^{-34} \text{ m}.$$

That is roughly 10^{19} times smaller than a proton. Diffraction needs an aperture comparable to λ , and no such aperture exists.

Think of it this way. The physics is identical to the electron case. Only the number is hopeless. Quantum mechanics does not switch off at large mass; its effects just become unmeasurable.

Problems 1.1–1.5

- Two routes have $|A_1|^2 = |A_2|^2 = 0.3$. Find the largest and smallest possible values of $|A_1 + A_2|^2$, and state the classical answer.
- Show that for $A_1 = re^{i\theta_1}$ and $A_2 = re^{i\theta_2}$,

$$|A_1 + A_2|^2 = 4r^2 \cos^2\left(\frac{\theta_1 - \theta_2}{2}\right).$$

- Estimate the temperature at which a 1 GHz qubit has under 1% thermal excitation.
- Find the de Broglie wavelength of an electron accelerated through 100 V. (Use $E = eV$ and $p = \sqrt{2mE}$.)
- Explain in two sentences why adding a second slit can make a point on the screen *darker*, and why this cannot happen for bullets.

2 Double Slit with Bullets and Waves

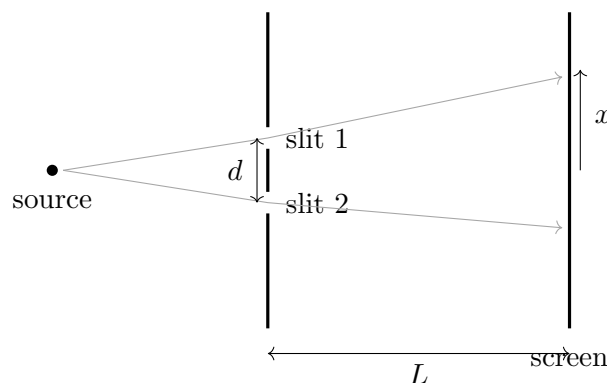
The Idea

We run one experiment three times, changing only what is fired: bullets, water waves, then electrons. The apparatus never changes.

Bullets behave one way. Waves behave a different way. Electrons do something that is neither, and *that* is the discovery.

This section establishes the two reference behaviours. The next section fires electrons.

2.1 The setup



Three curves are measured each time:

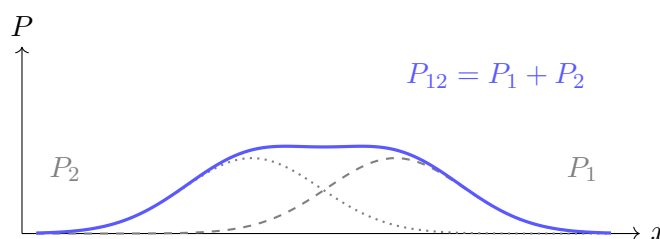
$$P_1(x) \text{ (slit 1 only), } P_2(x) \text{ (slit 2 only), } P_{12}(x) \text{ (both open).}$$

The only question in this section and the next: **how does P_{12} relate to P_1 and P_2 ?**

2.2 Run 1: bullets

Fire indestructible bullets from a gun on a shaky mount.

- Bullets arrive **one at a time, in lumps**. You never detect half a bullet.
- P_1 and P_2 are single broad humps behind each slit.
- With both slits open, $P_{12}(x) = P_1(x) + P_2(x)$.



Why

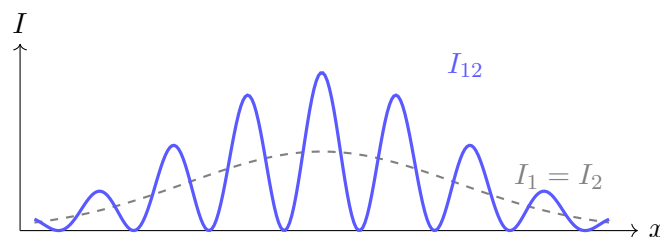
Each bullet went through slit 1 *or* slit 2. The two possibilities cannot both happen, so there is no way for them to combine, and we simply add their probabilities. This is ordinary probability, with nothing to explain.

Bullet behaviour: arrives in lumps, $P_{12} = P_1 + P_2$, no interference.

2.3 Run 2: water waves

Now a ripple tank. The detector measures **intensity**.

- Intensity arrives **continuously**, in any amount, never in lumps.
- I_1 and I_2 are broad humps as before.
- With both open there are **fringes**. At some positions the intensity is *zero* even though I_1 and I_2 are both nonzero there.



Amplitudes add, not intensities

A wave carries a complex amplitude h , and intensity is $I = |h|^2$. With both slits open the *amplitudes* add first, $h_{12} = h_1 + h_2$, so

$$I_{12} = |h_1 + h_2|^2 = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta,$$

where δ is the phase difference between the two paths.

The Interference Term and the Fringe Positions

$$I_{12} - (I_1 + I_2) = 2\sqrt{I_1 I_2} \cos \delta.$$

$\cos \delta = +1$ gives a bright band; $\cos \delta = -1$ gives a dark band.

The extra path from slit 2 is $d \sin \theta$, so

$$\text{bright: } d \sin \theta = m\lambda, \quad \text{dark: } d \sin \theta = \left(m + \frac{1}{2}\right) \lambda,$$

and for small angles ($\sin \theta \approx x/L$) the band spacing is

$$\Delta x = \frac{\lambda L}{d}$$

Wave behaviour: arrives continuously, $I_{12} \neq I_1 + I_2$, interference with fringes.

2.4 Worked examples

Example 2.1. Red light of wavelength 650 nm passes through two slits 0.2 mm apart onto a screen 1.5 m away. Find the fringe spacing.

Solution.

$$\Delta x = \frac{\lambda L}{d} = \frac{650 \times 10^{-9} \text{ m} \times 1.5 \text{ m}}{0.2 \times 10^{-3} \text{ m}} = \frac{9.75 \times 10^{-7}}{2 \times 10^{-4}} = 4.9 \times 10^{-3} \text{ m}.$$

So the bands sit about 4.9 mm apart, easily visible to the eye.

Example 2.2. Two slits pass unequal light, $I_1 = 4I_0$ and $I_2 = I_0$. Find $I_{\max}$, $I_{\min}$ and the fringe visibility $\mathcal{V} = (I_{\max} - I_{\min})/(I_{\max} + I_{\min})$. Comment on why the dark bands are not fully dark.

Solution.

Work with amplitudes, not intensities: $|h_1| = 2\sqrt{I_0}$ and $|h_2| = \sqrt{I_0}$.

$$I_{\max} = \left(2\sqrt{I_0} + \sqrt{I_0}\right)^2 = 9I_0, \quad I_{\min} = \left(2\sqrt{I_0} - \sqrt{I_0}\right)^2 = I_0.$$

$$\mathcal{V} = \frac{9I_0 - I_0}{9I_0 + I_0} = \frac{8}{10} = 0.8.$$

Why. Complete cancellation needs the two amplitudes to be equal in magnitude. Here one is twice the other, so the best they can do is partially cancel, and $I_{\min}$ stays above zero. This idea returns in Section 3 as the which-path trade-off.

Problems 2.1–2.5

1. Green light ($\lambda = 550 \text{ nm}$) gives fringes 3 mm apart on a screen 2 m away. Find the slit separation.
2. For $I_1 = I_2 = I_0$, show that $I_{\max} = 4I_0$, $I_{\min} = 0$, and find $\mathcal{V}$.
3. Using $\Delta x = \lambda L/d$, state what happens to the pattern if you (a) double d , (b) double λ , (c) double L .
4. Explain why $P_{12} = P_1 + P_2$ can never produce a dark band, whatever P_1 and P_2 look like.
5. A wave has $I_1 = 9I_0$ and $I_2 = 4I_0$. Find $I_{\max}$, $I_{\min}$ and $\mathcal{V}$.

3 Double Slit with Electrons and Probability Density

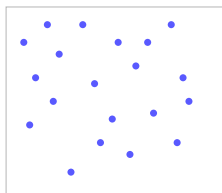
The Idea

Now fire electrons one at a time, so slowly that only one is ever inside the apparatus. They cannot be bumping into each other.

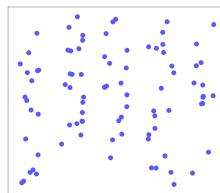
Two things happen, and together they break any classical picture:

- each electron lands at **one point**, like a bullet;
 - after many electrons the accumulated dots form the **wave** pattern, fringes and all.
- Electrons *arrive* like particles but *spread* like waves. The resolution is that the electron carries a wave-like amplitude which sets the *probability* of landing somewhere, and it is that amplitude which passes through both slits and interferes.

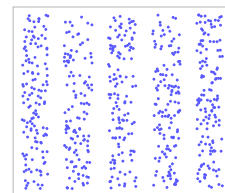
3.1 What the experiment shows



after 20 electrons



after 200



after 5000

Early on it looks random. The pattern emerges only from the statistics of many independent electrons. No single electron “makes a fringe”; the fringes live in the *probabilities*.

	Bullets	Waves	Electrons
Arrive in lumps?	yes	no	yes
Fringes?	no	yes	yes
$P_{12} = P_1 + P_2$?	yes	no	no

Electrons take the first row from bullets and the last two from waves.

Probability amplitude and probability density

Each electron has a complex **probability amplitude** $\psi(x)$. The chance of landing near x is

$$p(x) = |\psi(x)|^2, \quad \int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1.$$

With both slits open the *amplitudes* add:

$$\psi = \psi_1 + \psi_2 \quad \implies \quad p = |\psi_1|^2 + |\psi_2|^2 + 2 \operatorname{Re}(\psi_1^* \psi_2).$$

Since x is continuous, $|\psi(x)|^2$ is a probability *per unit length*, and the probability of landing in $[a, b]$ is $\int_a^b |\psi|^2 dx$.

Compare this with the water wave of Section 2. It is the same equation. Only the meaning of $|\cdot|^2$ changed: intensity for a wave, *probability* for an electron.

Density Is Not Probability

$|\psi(x)|^2$ is not itself a probability and may exceed 1. Only its integral over a region is a probability. For qubits everything is discrete and the distinction disappears, which is why the rest of the course is relaxed about it.

de Broglie Wavelength

$$\lambda = \frac{h}{p}.$$

Feed this into $\Delta x = \lambda L/d$ from Section 2 and you can predict electron fringe spacing directly.

3.2 Which slit did it go through?

Place a detector at the slits so you can tell which one each electron used.

The Result, and Why

With the which-path detector on, **the fringes vanish** and $P_{12} = P_1 + P_2$: exactly the bullet answer. Switch it off and the fringes return.

Interference requires the two routes to be **indistinguishable**. If anything anywhere records which slit was used, the amplitudes stop combining and you simply add probabilities.

It has nothing to do with a person looking. A photon bouncing off the electron and flying into a wall does the job perfectly well. What matters is whether a *record* exists. This is **decoherence**, and it is the main enemy of quantum computers.

Not About Consciousness

“Consciousness collapses the wavefunction” is not in the theory anywhere. Warm air molecules bumping into the electron destroy the fringes just as reliably as a physicist watching a screen.

You Cannot Have Both

Knowing the path and seeing fringes trade off against each other. Know the path completely and there are no fringes at all. Know nothing about the path and the fringes are as sharp as possible. Know a little and you get faint fringes.

If you want it as a formula, with $\mathcal{D}$ for how well the path is known and $\mathcal{V}$ for how sharp the fringes are,

$$\mathcal{D}^2 + \mathcal{V}^2 \leq 1.$$

The useful part is the sentence, not the formula: **it is a smooth trade-off, not an on/off switch.**

3.3 Worked examples

Example 3.1. Electrons are accelerated through 50 kV and sent through slits $1\text{ }\mu\text{m}$ apart onto a screen 1 m away. Find the de Broglie wavelength and the fringe spacing.

Solution.

Momentum from the kinetic energy $E = 50\text{ keV}$:

$$p = \sqrt{2mE} = \sqrt{2(9.11 \times 10^{-31})(50 \times 10^3)(1.6 \times 10^{-19})} \approx 1.2 \times 10^{-22}\text{ kg m/s}.$$

Wavelength:

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{1.2 \times 10^{-22}} \approx 5.5 \times 10^{-12}\text{ m} = 5.5\text{ pm}.$$

Fringe spacing:

$$\Delta x = \frac{\lambda L}{d} = \frac{5.5 \times 10^{-12} \times 1}{1 \times 10^{-6}} = 5.5 \times 10^{-6}\text{ m} = 5.5\text{ }\mu\text{m}.$$

The physical picture. λ is far smaller than an atom, yet the fringes are micrometres apart, because the geometry factor $L/d = 10^6$ magnifies them. These are roughly the numbers of the real Tonomura experiment.

Example 3.2. At a particular point on the screen the two slit amplitudes are $\psi_1 = 0.5$ and $\psi_2 = -0.5$. Find the probability density with only slit 1 open, only slit 2 open, and both open. Interpret the result.

Solution.

Only slit 1: $p_1 = |0.5|^2 = 0.25$.

Only slit 2: $p_2 = |-0.5|^2 = 0.25$.

Both open:

$$p_{12} = |0.5 + (-0.5)|^2 = |0|^2 = 0.$$

Why this matters. Opening a second slit took this point from 25% to 0%. *Blocking* a slit makes more electrons arrive here. That is impossible for bullets, where $P_{12} = P_1 + P_2 = 0.5$, and it is the single most striking fact in the whole experiment.

Problems 3.1–3.5

1. Find λ for an electron accelerated through 200 V, and the fringe spacing for $d = 0.5\text{ }\mu\text{m}$, $L = 0.8\text{ m}$.
2. At a point, $\psi_1 = 0.3$ and $\psi_2 = 0.3e^{i\pi/3}$. Find p_1 , p_2 and p_{12} . Is this point brighter or darker than the classical prediction?
3. For $\psi(x) = \sqrt{2/L} \sin(\pi x/L)$ on $[0, L]$, verify normalisation and find the probability of finding the particle in the left third.
4. A which-path detector achieves $\mathcal{D} = 0.6$. What is the largest visibility still possible?

5. Explain in three sentences, without using the word “observer”, why blowing warm air through the apparatus would kill the fringes.

4 Dirac Notation

The Idea

Dirac notation is a convenient way of writing vectors and matrices. There is no new physics in it.

Its advantage: it lets you name a state without committing to a basis, and it makes change-of-basis calculations nearly automatic. Once fluent, you will do in two lines what index notation takes half a page to do.

Four objects, and that is all:

- $|\psi\rangle$ (a **ket**) is a column vector;
- $\langle\psi|$ (a **bra**) is a row vector, conjugated;
- $\langle\phi|\psi\rangle$ is row times column, so a **number**;
- $|\phi\rangle\langle\psi|$ is column times row, so a **matrix**.

4.1 Kets, bras, and products

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad |\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

The label inside a ket is a name only. $|0\rangle$ is *not* the zero vector.

The bra is the conjugate transpose:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \implies \langle\psi| = \alpha^*\langle 0| + \beta^*\langle 1|.$$

The Conjugation Is Not Optional

If $|\psi\rangle = |0\rangle + i|1\rangle$ then $\langle\psi| = \langle 0| - i\langle 1|$. Dropping the conjugate is the single most common algebraic slip in this course, and it silently gives wrong probabilities rather than an obvious error.

Inner and outer products

Inner product (a number):

$$\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*, \quad \langle\psi|\psi\rangle = \|\psi\|^2 \geq 0.$$

A state is **normalised** when $\langle\psi|\psi\rangle = 1$.

Outer product (a matrix):

$$|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (|\phi\rangle\langle\psi|)|\chi\rangle = |\phi\rangle \underbrace{\langle\psi|\chi\rangle}_{\text{number}}.$$

Completeness: The Identity You Insert Everywhere

For any orthonormal basis $\{|e_i\rangle\}$,

$$\sum_i |e_i\rangle \langle e_i| = I$$

You may insert this anywhere in an expression. It is how you move between abstract states and components. For instance,

$$|\psi\rangle = I |\psi\rangle = \sum_i |e_i\rangle \langle e_i|\psi\rangle = \sum_i c_i |e_i\rangle, \quad c_i = \langle e_i|\psi\rangle,$$

so the expansion coefficients drop out with no work at all.

4.2 The operators you need

Adjoint, Hermitian, unitary

Adjoint: $A^\dagger$ means transpose and conjugate, $(A^\dagger)_{ij} = A_{ji}^*$.

$$(AB)^\dagger = B^\dagger A^\dagger, \quad (\alpha A)^\dagger = \alpha^* A^\dagger, \quad |\psi\rangle^\dagger = \langle\psi|.$$

Hermitian: $A = A^\dagger$. Real eigenvalues, orthogonal eigenvectors, so $A = \sum_k \lambda_k |a_k\rangle \langle a_k|$ with λ_k real. These represent **observables**.

Unitary: $U^\dagger U = U U^\dagger = I$, so $U^{-1} = U^\dagger$. Preserves inner products, hence total probability. These are the **gates**, and every one of them is reversible.

The Order Reverses

$(AB)^\dagger = B^\dagger A^\dagger$, *not* $A^\dagger B^\dagger$. Same pattern as $(AB)^{-1} = B^{-1} A^{-1}$. This appears in exam scripts every single year.

Two More Tools

Projector. $P = |\psi\rangle \langle\psi|$ for normalised $|\psi\rangle$, with $P^2 = P$ and $P^\dagger = P$. Measurements are built from projectors (Section 9).

Trace. The sum of the diagonal entries, with

$$\text{tr}(AB) = \text{tr}(BA), \quad \text{tr}(|\phi\rangle \langle\psi|) = \langle\psi|\phi\rangle.$$

Commutator. $[A, B] = AB - BA$. If $[A, B] = 0$ the two operators share eigenvectors, so both quantities can be sharp at once. If not, they cannot, and that is where the uncertainty principle comes from.

4.3 Worked examples

Example 4.1. Normalise $|v\rangle = 3|0\rangle + 4i|1\rangle$ and write down the corresponding bra.

Solution.

$$\langle v|v\rangle = (3)^*(3) + (4i)^*(4i) = 9 + (-4i)(4i) = 9 + 16 = 25.$$

So $\|v\| = 5$ and

$$|\tilde{v}\rangle = \frac{3}{5}|0\rangle + \frac{4i}{5}|1\rangle, \quad \langle\tilde{v}| = \frac{3}{5}\langle 0| - \frac{4i}{5}\langle 1|.$$

Watch the sign. $(4i)^*(4i) = (-4i)(4i) = -16i^2 = +16$, not -16 . Getting this backwards gives $\langle v|v\rangle = -7$, which should immediately tell you something is wrong since the norm cannot be negative.

Example 4.2. For $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}i|1\rangle$ and $|\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, compute $\langle\phi|\psi\rangle$ and the probability of finding $|\psi\rangle$ in the state $|\phi\rangle$.

Solution.

$$\langle\phi|\psi\rangle = \frac{1}{\sqrt{2}}(\langle 0| + \langle 1|)\left(\frac{1}{\sqrt{3}}|0\rangle + i\sqrt{\frac{2}{3}}|1\rangle\right) = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{3}} + i\sqrt{\frac{2}{3}}\right).$$

The probability is the modulus squared:

$$|\langle\phi|\psi\rangle|^2 = \frac{1}{2}\left(\frac{1}{3} + \frac{2}{3}\right) = \frac{1}{2}.$$

So there is a 50% chance.

In one sentence. The real and imaginary parts add in quadrature, which is why the i does not cancel anything here.

Example 4.3. Write $|0\rangle$ in the basis $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ using the completeness relation, and state the two measurement probabilities.

Solution.

Insert $I = |+\rangle\langle +| + |-\rangle\langle -|$:

$$|0\rangle = |+\rangle\langle +|0\rangle + |-\rangle\langle -|0\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle.$$

Probabilities: $|1/\sqrt{2}|^2 = \frac{1}{2}$ for each.

What to remember. A state that is perfectly definite in the Z basis is a perfect coin flip in the X basis. Note that no simultaneous equations were solved: inserting the identity did all the work.

Problems 4.1–4.5

1. Normalise $|v\rangle = (1 + i)|0\rangle + 2|1\rangle$ and write down $\langle v|$.
2. Compute $|+\rangle\langle +|$ as a 2×2 matrix and verify $P^2 = P$.
3. Verify $\text{tr}(|\phi\rangle\langle\psi|) = \langle\psi|\phi\rangle$ for $|\phi\rangle = |0\rangle$ and $|\psi\rangle = |+\rangle$.
4. Show that if A and B are both Hermitian then AB is Hermitian only when $[A, B] = 0$.
5. Verify that $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is both Hermitian and unitary, and that $H^2 = I$.

5 State of a Quantum System

The Idea

A quantum state is a normalised vector. That is the entire definition.

Two practical points follow, and both catch people out:

- multiplying the *whole* state by a phase changes nothing physical;
- changing the phase of *one part* relative to another changes everything.

Getting these two straight is the most useful thing in this section.

Qubit state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Measuring in the $\{|0\rangle, |1\rangle\}$ basis gives 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.

First, What Phase Actually Is

Every amplitude is a complex number, so it has a **size** and a **direction** in the complex plane. That direction is the phase.

size of an amplitude $\rightarrow$ how much of it there is

phase of an amplitude $\rightarrow$ how it combines with other amplitudes

Two amplitudes of the same size pointing the same way reinforce, $A + A = 2A$. Two pointing opposite ways cancel, $A - A = 0$. This is exactly the double slit of Section 2, now written for one qubit instead of two paths.

Global Phase Is Unobservable

$|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ are the **same physical state**, because every prediction has the form $|\langle\phi|\psi\rangle|^2$ or $\langle\psi|A|\psi\rangle$, and

$$|\langle\phi|e^{i\theta}\psi\rangle|^2 = |e^{i\theta}|^2 |\langle\phi|\psi\rangle|^2 = |\langle\phi|\psi\rangle|^2.$$

Global vs. Relative Phase

Global phase multiplies everything:

$$e^{i\pi/4} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \text{ is the same state as } \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

Relative phase sits between the parts:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

are **orthogonal**, hence perfectly distinguishable. Completely different states.

The test: pull out a common phase factor. Whatever is left over is relative, and it matters.

In the picture from the box above: a **global** phase rotates every amplitude together, so nothing about how they combine changes. A **relative** phase rotates one amplitude against another, which is precisely what decides whether they reinforce or cancel.

Superposition Is Not a Coin Flip

$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ gives a *certain* answer if you measure in the $|\pm\rangle$ basis. A genuine coin flip between $|0\rangle$ and $|1\rangle$ gives 50/50 in *every* basis. They are different physical situations, and Unit 2 makes the difference precise with density matrices.

5.1 Worked examples

Example 5.1. Decide whether each pair describes the same physical state. (a) $|0\rangle$ and $-|0\rangle$. (b) $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{i}{\sqrt{2}}(|0\rangle + |1\rangle)$. (c) $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$.

Solution.

(a) **Same.** $-|0\rangle = e^{i\pi}|0\rangle$, a global phase.

(b) **Same.** The factor i multiplies both terms, so it factors out entirely.

(c) **Different.** Here i multiplies only the $|1\rangle$ term and cannot be factored out. Confirm by computing the overlap:

$$|\langle +|\psi \rangle|^2 = \left| \frac{1}{4}(1 + i)^2 \right|^2 = \frac{1}{4}(2)^2 = \frac{1}{2} \neq 1.$$

If they were the same state the overlap would be 1.

Example 5.2. For $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}e^{i\pi/3}|1\rangle$, find $P(0)$ in the Z basis and $P(+)$ in the X basis. Comment on the role of the phase.

Solution.

Z basis. $P(0) = |1/\sqrt{2}|^2 = \frac{1}{2}$ and likewise $P(1) = \frac{1}{2}$. The phase does not appear.

X basis.

$$\langle +|\psi\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}e^{i\pi/3}\right) = \frac{1}{2}(1 + e^{i\pi/3}),$$

$$P(+)=\frac{1}{4}|1+e^{i\pi/3}|^2=\frac{1}{4}(2+2\cos\frac{\pi}{3})=\frac{1}{4}(2+1)=0.75.$$

Why this matters. The phase was completely invisible in one basis and decisive in another. That is relative phase doing its job, and it is why choosing the measurement basis matters.

Problems 5.1–5.5

1. For $|\psi\rangle = \frac{1}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle$, find $P(0)$ and $P(1)$.
2. Same or different: (a) $|1\rangle$ and $i|1\rangle$; (b) $|0\rangle + |1\rangle$ and $|0\rangle - |1\rangle$; (c) $i|0\rangle + i|1\rangle$ and $|0\rangle + |1\rangle$.
3. For $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}|1\rangle)$, find $P(+)$ as a function of φ and sketch it.
4. Find β , up to phase, so that $|\psi\rangle = 0.6|0\rangle + \beta|1\rangle$ is normalised.
5. Show that $P(+)+P(-)=1$ for any normalised state.

6 Hilbert Space Representation

The Idea

A Hilbert space is “a vector space where you can take inner products.” Nothing exotic. For this course it means $\mathbb{C}^n$ with the usual dot product, conjugated on the left.

The one fact to take away: when you combine n qubits the dimensions **multiply**, they do not add. Two qubits need 4 numbers, three need 8, ten need 1024. That exponential growth is both why quantum computers are powerful and why simulating them is hard.

Hilbert space, practically

A complex vector space $\mathcal{H}$ with an inner product $\langle\phi|\psi\rangle$. For us, $\mathcal{H} = \mathbb{C}^n$.

System	Space	Dimension
one qubit	$\mathbb{C}^2$	2
two qubits	$\mathbb{C}^2 \otimes \mathbb{C}^2$	4
n qubits	$(\mathbb{C}^2)^{\otimes n}$	2^n
particle on a line	functions $\psi(x)$	infinite

6.1 Combining systems: the tensor product

If A is in $|\psi_A\rangle$ and B is in $|\psi_B\rangle$, the pair is in $|\psi_A\rangle \otimes |\psi_B\rangle$, usually written $|\psi_A\rangle |\psi_B\rangle$. For two qubits the four basis states are $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, and a general state is

$$|\psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle, \quad \sum |c_{ij}|^2 = 1.$$

Dimensions Multiply

n classical bits: you track n values.

n qubits: you track 2^n amplitudes.

For $n = 50$ that is over 10^{15} complex numbers, roughly 18 petabytes at double precision. This is exactly why a classical computer cannot simulate a large quantum computer, and why the hardware is worth building.

Product test for two qubits

Not every joint state factorises. The state $c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle$ is a **product state** if and only if

$$c_{00}c_{11} - c_{01}c_{10} = 0.$$

Otherwise it is **entangled**, which is Unit 3.

6.2 Worked examples

Example 6.1. Form the joint state of $|\psi_A\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|\psi_B\rangle = |1\rangle$, and confirm with the product test that it is not entangled.

Solution.

$$|\psi_A\rangle \otimes |\psi_B\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle).$$

Reading off coefficients: $c_{01} = c_{11} = 1/\sqrt{2}$ and $c_{00} = c_{10} = 0$.

$$c_{00}c_{11} - c_{01}c_{10} = 0 - 0 = 0.$$

Product state, as expected, since we built it from two independent pieces.

Example 6.2. Apply the product test to $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and to $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.

Solution.

First state. $c_{00} = c_{11} = 1/\sqrt{2}$, $c_{01} = c_{10} = 0$:

$$c_{00}c_{11} - c_{01}c_{10} = \frac{1}{2} - 0 = \frac{1}{2} \neq 0 \implies \text{entangled.}$$

This is the Bell state, built in Section 12.

Second state. All four coefficients equal $\frac{1}{2}$:

$$c_{00}c_{11} - c_{01}c_{10} = \frac{1}{4} - \frac{1}{4} = 0 \implies \text{product.}$$

Indeed it factorises as $|+\rangle \otimes |+\rangle$.

What to remember. “Looks complicated” and “is entangled” are different things. Always run the test.

Problems 6.1–6.5

1. Write out $|+\rangle \otimes |+\rangle$ in the $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ basis.
2. Apply the product test to $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$.
3. Apply the product test to $\frac{1}{\sqrt{3}}|00\rangle + \sqrt{\frac{2}{3}}|11\rangle$.
4. How many real numbers describe an n -qubit pure state after imposing normalisation and ignoring global phase? Evaluate for $n = 1, 2, 10$.
5. A hard drive stores 1 TB. What is the largest n for which a general n -qubit state fits, at 16 bytes per amplitude?

7 Quantum Mathematics: Linearity of States and Operators

The Idea

A gate does not care whether its input is simple or complicated.

Suppose you know only two things about a gate U : what it does to $|0\rangle$, and what it does to $|1\rangle$. That turns out to be enough to know what it does to *every* state, because

$$U(\alpha|0\rangle + \beta|1\rangle) = \alpha U|0\rangle + \beta U|1\rangle.$$

The gate acts on each piece separately and leaves the pieces added together. That property is called **linearity**, and every gate in quantum mechanics has it.

This is good news and bad news. The good news is that a 2×2 matrix is enough to describe a gate acting on infinitely many states. The bad news is that linearity also **rules things out**, and the most famous casualty is copying. We build that argument in Section 7.1.

Linearity

Of states. If $|\psi_1\rangle$ and $|\psi_2\rangle$ are allowed, so is any normalised $\alpha|\psi_1\rangle + \beta|\psi_2\rangle$.

Of operators. Every operator satisfies

$$U(\alpha|\psi_1\rangle + \beta|\psi_2\rangle) = \alpha U|\psi_1\rangle + \beta U|\psi_2\rangle.$$

The Practical Consequence

To specify a one-qubit gate you only need to say what it does to $|0\rangle$ and $|1\rangle$. Everything else follows automatically. That is why gates are given as small matrices, and why a 2×2 matrix can describe the action on a continuum of states.

7.1 Can we build a quantum photocopier?

Here is a machine that sounds completely reasonable. You hand it a qubit in some state $|\psi\rangle$ and a blank qubit in $|0\rangle$, and it hands you back two copies:

$$|\psi\rangle |0\rangle \xrightarrow{U} |\psi\rangle |\psi\rangle.$$

Let us try to build it. Start with the easy cases. Copying $|0\rangle$ and copying $|1\rangle$ is no trouble at all, and in fact you already know a gate that does it: CNOT.

$$U |0\rangle |0\rangle = |0\rangle |0\rangle, \quad U |1\rangle |0\rangle = |1\rangle |1\rangle.$$

So far so good. Now try a superposition. If the machine can copy $|0\rangle$ and it can copy $|1\rangle$, surely it can copy $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$?

It cannot, and linearity is what stops it.

Watch What Goes Wrong

What the machine actually does. Linearity leaves us no choice. The gate acts on each piece and keeps them added:

$$U |+\rangle |0\rangle = \frac{1}{\sqrt{2}}(U |0\rangle |0\rangle + U |1\rangle |0\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

What we asked for. Two independent copies of $|+\rangle$:

$$|+\rangle |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$

These are not the same state. The first has no $|01\rangle$ or $|10\rangle$ at all; the second has all four terms equally. So the machine does not copy $|+\rangle$, and no adjustment will fix it, because linearity was never optional.

No-cloning theorem

No gate can copy an arbitrary unknown quantum state. Formally: there is no unitary U with $U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle$ for every $|\psi\rangle$.

Notice that the machine did not fail by producing garbage. It produced $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, which is a perfectly good state, and one you will see constantly. It is entangled. Instead of two copies, you get two qubits that are linked.

Why You Should Care

- You cannot back up a qubit the way you back up a file. Quantum error correction has to work around this, and it does (Unit 3).
- An eavesdropper cannot copy a quantum message and pass the original along undetected. That is what makes quantum key distribution secure.
- Note what is *not* forbidden: copying a state you already *know*, or copying from a known orthogonal set. Only *arbitrary unknown* states are protected.

7.2 Worked examples

Example 7.1. Given only that $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$, find $H|+\rangle$ without using any matrices.

Solution.

$$H|+\rangle = H\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) = \frac{1}{\sqrt{2}}(H|0\rangle + H|1\rangle) = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle).$$

Substituting the definitions of $|\pm\rangle$:

$$= \frac{1}{\sqrt{2}}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} + \frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = \frac{1}{2}(2|0\rangle) = |0\rangle.$$

So $H|+\rangle = |0\rangle$, consistent with $H^2 = I$.

Example 7.2. A student proposes a gate N that “strips the phases”, acting as $N(\alpha|0\rangle + \beta|1\rangle) = |\alpha||0\rangle + |\beta||1\rangle$. Show that N cannot be a quantum gate.

Solution.

Test linearity on $|0\rangle - |1\rangle$ (ignoring normalisation).

Applying N directly:

$$N(|0\rangle - |1\rangle) = |1||0\rangle + |-1||1\rangle = |0\rangle + |1\rangle.$$

Applying N termwise, as linearity would require:

$$N|0\rangle - N|1\rangle = |0\rangle - |1\rangle.$$

These disagree, so N is not linear, and therefore not unitary, and therefore not a legal quantum gate.

Think of it this way. Any operation that discards relative phase is non-linear. That is why decoherence is described by a different formalism (a channel, Unit 2) and not by a gate.

Problems 7.1–7.5

1. Given $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$, find $X|+\rangle$ and $X|-\rangle$ using linearity only.
2. Given $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$, find $Z|+\rangle$.
3. Adapt the no-cloning argument to show that no unitary can perform $|\psi\rangle|\psi\rangle \rightarrow |\psi\rangle|0\rangle$ for arbitrary $|\psi\rangle$.
4. Is “measure in the Z basis and output the result” a linear operation? Explain.
5. Show that a machine able to clone both $|0\rangle$ and $|+\rangle$ leads to a contradiction.

8 Discrete and Continuous Base Representations

The Idea

A basis is just **the list of possible answers** to a measurement question.

Ask “is it 0 or 1?” and the answers are $|0\rangle$ and $|1\rangle$. Ask a different question and you get a different list. Same state, different question, different list of answers.

Two kinds of list:

- countable answers (0 or 1, up or down) $\Rightarrow$ **discrete** basis, and you write sums;
- a continuum of answers (a position can be any real number) $\Rightarrow$ **continuous** basis, and you write integrals.

That is the only difference. The rest of this course uses discrete bases.

What makes a list a basis

Two conditions, in plain terms:

- **Orthonormal** — the answers do not overlap, and each has length 1: $\langle e_i | e_j \rangle = \delta_{ij}$.
 - **Complete** — the answers cover everything, nothing is left out: $\sum_i |e_i\rangle \langle e_i| = I$.
- Then any state splits over the answers as $|\psi\rangle = \sum_i c_i |e_i\rangle$ with $c_i = \langle e_i | \psi \rangle$, and the probabilities $|c_i|^2$ add to 1.

8.1 Three qubit bases you will use constantly

Basis	The two answers	The question it asks
Z (computational)	$ 0\rangle, 1\rangle$	is it 0 or 1?
X (Hadamard)	$ \pm\rangle = \frac{1}{\sqrt{2}}(0\rangle \pm 1\rangle)$	is it a + or a – superposition?
Y (circular)	$ \pm i\rangle = \frac{1}{\sqrt{2}}(0\rangle \pm i 1\rangle)$	same, but with an i in the phase

The One Fact To Remember

A state that gives a **certain** answer in one of these three bases is a **perfect coin flip** in the other two.

$|0\rangle$ is certain in Z . Measure it in X and you get 50/50. That is all “choosing a basis” means in practice: you pick the question, and the state decides how sharp the answer is.

8.2 Continuous bases, briefly

For a particle on a line the answer to “where is it?” can be any real number x . So the list of answers is a continuum, written $|x\rangle$.

Nothing conceptual changes. Two symbols get swapped:

	Discrete	Continuous
add up the answers	$\sum_i$	$\int dx$
coefficients	$c_i = \langle e_i \psi \rangle$	$\psi(x) = \langle x \psi \rangle$
total probability	$\sum_i c_i ^2 = 1$	$\int \psi(x) ^2 dx = 1$

So What Is the Wavefunction?

$$\psi(x) = \langle x | \psi \rangle$$

The wavefunction is **just the list of coefficients**, one for each position. It is the same c_i you have been writing all along, with a continuous index instead of a counting index.

If that is the only thing you take from this section, that is enough.

$|x\rangle$ Is a Convenience, Not a State

A particle cannot sit at exactly one point, so $|x\rangle$ is not a physical state. It is a bookkeeping device that makes the integrals work. Real particles are spread out over a range of x . The same goes for momentum states $|p\rangle$.

You will not need to compute with $|x\rangle$ in this course.

8.3 Worked examples

Example 8.1. Write $|1\rangle$ in the $|\pm\rangle$ basis and give the two measurement probabilities.

Solution.

Split $|1\rangle$ over the two answers of the X basis:

$$|1\rangle = |+\rangle \langle + | 1 \rangle + |-\rangle \langle - | 1 \rangle = \frac{1}{\sqrt{2}} |+\rangle - \frac{1}{\sqrt{2}} |-\rangle.$$

Probabilities: $|\pm 1/\sqrt{2}|^2 = \frac{1}{2}$ each.

What to remember. $|1\rangle$ is a completely certain answer in the Z basis and a coin flip in the X basis. Nothing about the state changed; only the question did.

Example 8.2. A particle is equally likely to be anywhere between $x = 0$ and $x = L$, so $\psi(x) = A$ on that range and zero outside. Find A and the probability of finding the particle in the left half.

Solution.

Total probability must be 1:

$$\int_0^L |A|^2 dx = |A|^2 L = 1 \quad \implies \quad A = \frac{1}{\sqrt{L}}.$$

Left half:

$$\int_0^{L/2} \frac{1}{L} dx = \frac{1}{L} \cdot \frac{L}{2} = \frac{1}{2}.$$

In one sentence. Exactly what you would guess. The integral is doing the same job as adding up $|c_i|^2$ in the discrete case.

Problems 8.1–8.5

1. Write $|0\rangle$ in the $|\pm\rangle$ basis and give the two probabilities.
2. Write $|+\rangle$ in the $\{|+\rangle, |-\rangle\}$ basis and give the two probabilities.
3. Verify that $|+\rangle$ and $|-\rangle$ are orthonormal.
4. Show that $|\langle + | + \rangle|^2 = \frac{1}{2}$, i.e. a state certain in the X basis is a coin flip in the Y basis.
5. For $\psi(x) = A$ on $0 < x < L$, find the probability of being in the middle third.

9 Basic Postulates of Quantum Mechanics

The Idea

Everything so far was motivation. Now the theory itself.

Do not think of it as four laws handed down to be memorised. Think of it as **four questions any physical theory has to answer**, and quantum mechanics' answers to them.

The question	Quantum mechanics says
What is the system?	A normalised vector $ \psi\rangle$.
How does it change on its own?	A unitary matrix acts on it, $ \psi'\rangle = U \psi\rangle$.
What happens when we look?	We get one answer at random, with probability $ \langle a_m \psi \rangle ^2$, and the state jumps to that answer.
How do we put two systems together?	Tensor product: list all combinations of their states.

Read the table first. The formal statements below say the same four things precisely.

Postulate 1: State

A physical system is described by a Hilbert space $\mathcal{H}$. The state is a **unit vector** $|\psi\rangle \in \mathcal{H}$, defined up to a global phase.

Postulate 2: Evolution

A closed system evolves by a **unitary** matrix, $|\psi(t_2)\rangle = U |\psi(t_1)\rangle$. In continuous time this is the Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle, \quad U(t) = e^{-iHt/\hbar},$$

with H the Hermitian Hamiltonian (the energy operator).

Why Unitary

Unitary means “preserves lengths.” The length of the state vector is the total probability. Non-unitary evolution would let probability leak away or blow up.

So: H Hermitian (real energies) $\Rightarrow U$ unitary (probability conserved). And since $U^{-1} = U^\dagger$ always exists, quantum evolution is always **reversible**.

A trick you will use constantly: if $A^2 = I$ (true for every Pauli), then

$$e^{-i\theta A} = \cos \theta I - i \sin \theta A.$$

Postulate 3: Measurement

Pick a basis, i.e. pick the question you are asking. Measuring $|\psi\rangle$ in the basis $\{|a_m\rangle\}$ gives answer m with probability

$$p(m) = |\langle a_m | \psi \rangle|^2, \quad \text{and the state becomes } |a_m\rangle.$$

This is the **Born rule**, and it is the only version you need in this course.

An **observable** is a Hermitian operator $A = \sum_m \lambda_m |a_m\rangle \langle a_m|$: it packages the measurement basis together with a number λ_m to report for each answer. Its average over many runs is

$$\langle A \rangle = \langle \psi | A | \psi \rangle = \sum_m \lambda_m p(m).$$

Measurement Is Destructive

After measuring $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ and getting 0, the state is exactly $|0\rangle$. The β part is gone permanently. You cannot measure again to learn β , and you cannot undo it. This is why a qubit yields only **one bit**, even though α and β between them contain infinitely many.

Uncertainty

With $\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$,

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|.$$

If $[A, B] = 0$ the bound is zero and both can be sharp. If not, they cannot. For

position and momentum $[\hat{x}, \hat{p}] = i\hbar$ gives $\Delta x \Delta p \geq \hbar/2$; for a qubit $[X, Z] = -2iY$.

What Uncertainty Does Not Say

It bounds the *spread of results over many runs* on identically prepared systems. It does not say a single measurement is imprecise. Do not write “measuring position disturbs momentum, therefore $\Delta x \Delta p \geq \hbar/2$ ” in an exam; that is a slogan, not the theorem.

Postulate 4: Composite systems

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n.$$

If the parts are prepared in $|\psi_1\rangle, \dots, |\psi_n\rangle$, the whole is $|\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$. Note the word “if”: the space contains far more than products, and the extras are the entangled states.

9.1 Worked examples

Example 9.1. A qubit is driven by $H = \frac{\hbar\Omega}{2}X$ starting from $|0\rangle$. Find $|\psi(t)\rangle$ and $P(1)$, and explain how this is used to build an X gate in hardware.

Solution.

Since $X^2 = I$, use the exponential trick with $\theta = \Omega t/2$:

$$U(t) = e^{-i\Omega t X/2} = \cos \frac{\Omega t}{2} I - i \sin \frac{\Omega t}{2} X.$$

Applying to $|0\rangle$:

$$|\psi(t)\rangle = \cos \frac{\Omega t}{2} |0\rangle - i \sin \frac{\Omega t}{2} |1\rangle, \quad P(1) = \sin^2 \frac{\Omega t}{2}.$$

Hardware reading. The population oscillates between $|0\rangle$ and $|1\rangle$ at the Rabi frequency Ω . Holding the drive for $t = \pi/\Omega$ gives $P(1) = 1$, i.e. an X gate up to global phase. Half that time gives an equal superposition. This is literally how single-qubit gates are implemented: a resonant pulse of calibrated duration.

Example 9.2. For $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$, compute $\langle Z \rangle$, ΔZ and $\langle X \rangle$.

Solution.

Probabilities: $p(0) = \frac{1}{3}$, $p(1) = \frac{2}{3}$.

$$\langle Z \rangle = (+1)\frac{1}{3} + (-1)\frac{2}{3} = -\frac{1}{3}.$$

Since $Z^2 = I$, $\langle Z^2 \rangle = 1$, so

$$\Delta Z = \sqrt{1 - \frac{1}{9}} = \sqrt{\frac{8}{9}} \approx 0.943.$$

For X , use $\langle X \rangle = 2 \operatorname{Re}(\alpha^* \beta)$ with both coefficients real and positive:

$$\langle X \rangle = 2 \cdot \frac{1}{\sqrt{3}} \cdot \sqrt{\frac{2}{3}} = 2\sqrt{\frac{2}{9}} = \frac{2\sqrt{2}}{3} \approx 0.943.$$

Sanity check. $\langle X \rangle^2 + \langle Z \rangle^2 = \frac{8}{9} + \frac{1}{9} = 1$, consistent with a pure state having $|\vec{r}| = 1$ (Section 11).

Problems 9.1–9.5

1. For $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$, find $\langle Z \rangle$, $\langle X \rangle$ and ΔZ .
2. Take $H = \frac{\hbar\omega}{2}Z$. Find $U(t)$ and apply it to $|+\rangle$. What changes and what does not?
3. Show that $\sum_m p(m) = 1$ for any normalised state measured in any orthonormal basis.
4. A qubit is in $|+\rangle$. You measure Z , get 0, then measure X . Give the probabilities of the second measurement, and explain why the answer is not “+ with certainty”.
5. Show that if $[A, H] = 0$ then $\langle A \rangle$ is constant in time.

10 Matrix Representation of State Vectors

The Idea

Time to put numbers in. Fix a basis, write states as columns and operators as matrices, and everything becomes concrete arithmetic.

The main content is the three **Pauli matrices**. They are the alphabet of qubit mathematics: every 2×2 matrix is built from them, every gate is built from them, and every single-qubit measurement is one of them. Learn the multiplication table properly and the rest of the course gets noticeably easier.

Before the entries, learn what the three of them *do*.

What Each Pauli Does

- X **flips** the qubit: $|0\rangle \leftrightarrow |1\rangle$. It is the quantum NOT gate.
- Z leaves $|0\rangle$ and $|1\rangle$ alone but **flips the sign** of $|1\rangle$. It changes relative phase, so it does nothing visible in the Z basis and everything in the X basis.
- Y does **both at once**: a flip and a phase change. In fact $Y = iXZ$.
 X is a bit flip, Z is a phase flip, Y is both. Now the matrices will look like they mean something.

The Pauli matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Read X off the matrix: it swaps the two entries of the column vector, i.e. it swaps $|0\rangle$ and $|1\rangle$. Read Z off: it leaves the top entry and negates the bottom one.

Pauli Algebra: Memorise This

$$X^2 = Y^2 = Z^2 = I$$

$$XY = iZ, \quad YZ = iX, \quad ZX = iY \quad (\text{cyclic: } X \rightarrow Y \rightarrow Z \rightarrow X)$$

$$YX = -iZ, \quad ZY = -iX, \quad XZ = -iY \quad (\text{reverse the order, pick up a minus})$$

Every Pauli is Hermitian, unitary, traceless, with eigenvalues $+1$ and -1 . Two *different* Paulis **anticommute**: $XZ = -ZX$.

	eigenvalue $+1$	eigenvalue -1
Z	$ 0\rangle$	$ 1\rangle$
X	$ +\rangle = \frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$	$ -\rangle = \frac{1}{\sqrt{2}}(0\rangle - 1\rangle)$
Y	$ +i\rangle = \frac{1}{\sqrt{2}}(0\rangle + i 1\rangle)$	$ -i\rangle = \frac{1}{\sqrt{2}}(0\rangle - i 1\rangle)$

Every 2×2 matrix is a Pauli combination

$$M = c_0 I + c_x X + c_y Y + c_z Z, \quad c_\mu = \frac{1}{2} \text{tr}(\sigma_\mu M).$$

Four cases worth knowing cold:

$$|0\rangle\langle 0| = \frac{I + Z}{2}, \quad |1\rangle\langle 1| = \frac{I - Z}{2}, \quad |\pm\rangle\langle \pm| = \frac{I \pm X}{2}, \quad |\pm i\rangle\langle \pm i| = \frac{I \pm Y}{2}.$$

10.1 Common gates

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

Hadamard: The Most Used Gate

$$H = \frac{X + Z}{\sqrt{2}}, \quad H^2 = I, \quad H = H^\dagger,$$

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle, \quad H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle,$$

$$HXH = Z, \quad HZH = X.$$

H swaps bit flips and phase flips. Also $S^2 = Z$ and $T^2 = S$.

10.2 Two qubits

Tensor product of matrices, and CNOT

$$A \otimes B = \begin{pmatrix} A_{11}B & A_{12}B \\ A_{21}B & A_{22}B \end{pmatrix}, \quad (A \otimes B)(C \otimes D) = (AC) \otimes (BD).$$

Ordering the basis as $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad |a, b\rangle \mapsto |a, a \oplus b\rangle,$$

that is, “flip the second qubit if the first is 1.”

Qiskit Bit Ordering

Qiskit is **little-endian**: in an output string such as ‘01’, the *rightmost* character is qubit 0. These notes and most textbooks put qubit 0 on the left. If your simulator output looks bit-reversed against your hand calculation, this is why. Habit: print the statevector for a trivial circuit first and confirm the ordering before trusting anything.

10.3 Worked examples

Example 10.1. Verify $HXH = Z$ by direct matrix multiplication.

Solution.

$$HXH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Take the right-hand pair first:

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Then

$$HXH = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z.$$

The physical picture. Conjugating by H turns a bit flip into a phase flip. That is why Hadamards bracket so many circuits: they move a problem into whichever basis makes it easy.

Example 10.2. Starting from $|00\rangle$, apply H to qubit 0 and then CNOT. Give the resulting state and confirm with the product test that it is entangled.

Solution.

Step 1, Hadamard on the first qubit:

$$|00\rangle \longrightarrow \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

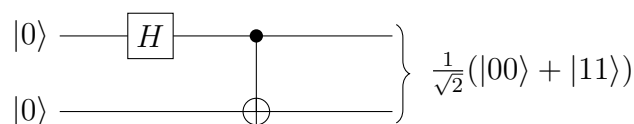
Step 2, CNOT flips the second qubit only where the first is 1, so $|10\rangle$ becomes $|11\rangle$ while $|00\rangle$ is untouched:

$$\longrightarrow \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Product test: $c_{00} = c_{11} = 1/\sqrt{2}$, $c_{01} = c_{10} = 0$, so

$$c_{00}c_{11} - c_{01}c_{10} = \frac{1}{2} \neq 0 \implies \text{entangled.}$$

Why this matters. Two gates, and the state is no longer a product. This circuit is the workhorse of the entire course.

**Problems 10.1–10.5**

1. Verify $XY = iZ$ by direct matrix multiplication.
2. Show that $|-\rangle\langle-| = (I - X)/2$.
3. Write $X \otimes Z$ as a 4×4 matrix and check that it is unitary.
4. Apply CNOT to $|+0\rangle$ and to $|+1\rangle$. Are the results entangled?
5. Compute $H \otimes H$ applied to $|00\rangle$ and express the answer in the computational basis.

11 Geometry of Probability Distributions: The Bloch Sphere

The Idea

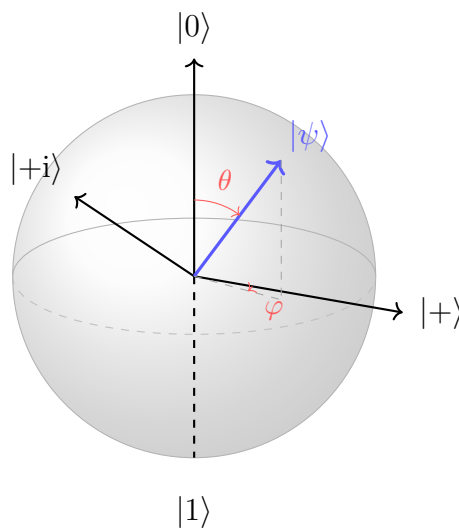
A classical bit's state is one number p between 0 and 1. Draw it and you get a **line segment**, with the two definite states at the ends.

A qubit's state needs two numbers. Draw it and you get a **sphere**, with every surface point a different pure state.

That is the whole idea: the classical bit's segment becomes the qubit's sphere. It is worth learning because gates become *rotations*, and rotations are far easier to picture than 2×2 complex matrices.

11.1 A map of every qubit state

Rather than staring at two complex numbers α and β , we can draw a pure qubit as a **single point on a sphere**. Here are the landmarks.



State	Written out	(θ, φ)	Position
$ 0\rangle$	$ 0\rangle$	$(0, -)$	north pole
$ 1\rangle$	$ 1\rangle$	$(\pi, -)$	south pole
$ +\rangle$	$\frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$	$(\frac{\pi}{2}, 0)$	$+x$
$ -\rangle$	$\frac{1}{\sqrt{2}}(0\rangle - 1\rangle)$	$(\frac{\pi}{2}, \pi)$	$-x$
$ +i\rangle$	$\frac{1}{\sqrt{2}}(0\rangle + i 1\rangle)$	$(\frac{\pi}{2}, \frac{\pi}{2})$	$+y$
$ -i\rangle$	$\frac{1}{\sqrt{2}}(0\rangle - i 1\rangle)$	$(\frac{\pi}{2}, \frac{3\pi}{2})$	$-y$

The Payoff

Once every state is a point on this sphere, **every gate becomes a rotation** of that point.

X , Y and Z each spin the sphere by 180° about their own axis. S turns it 90° about z ; T turns it 45° . That is far easier to picture than multiplying 2×2 complex matrices, and it is why the sphere is worth learning.

11.2 Counting the parameters

two complex numbers α, β	4 real numbers
normalisation $ \alpha ^2 + \beta ^2 = 1$	-1
global phase is unobservable	-1
left over	2

Two angles. Two angles is a sphere.

Standard form of a qubit state

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad \theta \in [0, \pi], \quad \varphi \in [0, 2\pi).$$

θ is the angle down from the north pole; φ is the angle around the equator. The **Bloch vector** is

$$\vec{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle) = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta),$$

and $|\vec{r}| = 1$ for every pure state, so pure states sit on the *surface*.

The Half-Angle, and Why Everyone Trips On It

Orthogonal states sit **opposite each other** on the sphere, not at right angles. $|0\rangle$ and $|1\rangle$ are orthogonal and they are 180° apart, at the two poles.

That is where the $\theta/2$ comes from: an angle of θ on the *sphere* corresponds to $\theta/2$ in the *state*.

Useful formula:

$$|\langle \psi | \phi \rangle|^2 = \frac{1 + \vec{r}_\psi \cdot \vec{r}_\phi}{2}.$$

Opposite points give overlap 0. The same point gives 1. **Perpendicular** Bloch vectors give $\frac{1}{2}$, not 0.

Gates Are Rotations

$$R_{\hat{n}}(\theta) = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (\hat{n} \cdot \vec{\sigma})$$

rotates the Bloch vector by θ about the axis $\hat{n}$.

Gate	Axis	Angle
X	x	180°
Y	y	180°
Z	z	180°
S	z	90°
T	z	45°
H	diagonal, $(x + z)/\sqrt{2}$	180°

No Bloch Sphere for Two Qubits

The picture works only for a single qubit. Drawing two spheres side by side shows each qubit separately and throws away the entanglement, which is exactly the interesting part. Keep this in mind for Unit 3.

11.3 Worked examples

Example 11.1. Find (θ, φ) and the Bloch vector for $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$.

Solution.

Match to the standard form. From the $|0\rangle$ coefficient,

$$\cos \frac{\theta}{2} = \frac{1}{2} \implies \frac{\theta}{2} = 60^\circ \implies \theta = 120^\circ.$$

Check the $|1\rangle$ coefficient: $\sin 60^\circ = \sqrt{3}/2$. ✓ It is real and positive, so $\varphi = 0$.

Bloch vector:

$$\vec{r} = (\sin 120^\circ \cos 0, \sin 120^\circ \sin 0, \cos 120^\circ) = (0.866, 0, -0.5).$$

Check the length: $0.866^2 + 0.5^2 = 0.75 + 0.25 = 1$. ✓

The physical picture. The state sits below the equator, tilted toward $+x$, which matches $P(1) = 3/4 > P(0)$.

Example 11.2. Use the Bloch geometry to explain why H maps $|0\rangle$ to $|+\rangle$, and verify that $Z|+\rangle = |-\rangle$ both geometrically and algebraically.

Solution.

Hadamard. H is a 180° rotation about the diagonal axis halfway between x and z . A 180° rotation about that axis **swaps** the x and z axes. So the north pole ($|0\rangle$, at $+z$) is carried to $+x$, which is $|+\rangle$. This is the geometric content of $HZH = X$ from Section 10.

Z on $|+\rangle$, geometrically. Z is a 180° rotation about the vertical axis. $|+\rangle$ sits at $+x$; rotating 180° about z carries $+x$ to $-x$, which is $|-\rangle$.

Z on $|+\rangle$, algebraically.

$$Z|+\rangle = \frac{1}{\sqrt{2}}(Z|0\rangle + Z|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle. \quad \checkmark$$

Problems 11.1–11.5

1. Find (θ, φ) and $\vec{r}$ for $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$.
2. Find the state at $\theta = 60^\circ$, $\varphi = 180^\circ$.
3. Using $|\langle\psi|\phi\rangle|^2 = (1 + \vec{r}_\psi \cdot \vec{r}_\phi)/2$, find the overlap of $|0\rangle$ with a state at polar angle θ , and confirm it equals $\cos^2(\theta/2)$.
4. Where does T send $|+\rangle$? Give the Bloch vector.
5. Apply X then Z to $|0\rangle$ using rotations, and check the answer algebraically.

12 Python Implementation of Basic Quantum Examples

The Idea

Everything in this unit can be checked in a few lines of Python, and you should check it. Typing the code and watching a number come out fixes ideas far better than re-reading.

Five short labs, one per big idea: the Pauli algebra, measurement, the Bloch sphere, interference, and entanglement. Each is self-contained. All of it is in `QEQC_Unit01_Lab.ipynb`.

12.1 Setup

```
# pip install qiskit qiskit-aer numpy
import numpy as np
from qiskit import QuantumCircuit, transpile
from qiskit.quantum_info import Statevector
from qiskit_aer import AerSimulator

np.set_printoptions(precision=4, suppress=True)
```

12.2 Lab 1: states and the Pauli algebra

```
ket0 = np.array([1, 0], dtype=complex)
ket1 = np.array([0, 1], dtype=complex)

I = np.eye(2, dtype=complex)
X = np.array([[0, 1], [1, 0]], dtype=complex)
Y = np.array([[0, -1j], [1j, 0]], dtype=complex)
Z = np.array([[1, 0], [0, -1]], dtype=complex)
H = (X + Z) / np.sqrt(2)

plus, minus = H @ ket0, H @ ket1

# check Section 10 instead of trusting it
for name, P in [("X", X), ("Y", Y), ("Z", Z)]:
    print(name, "squares to I ?", np.allclose(P @ P, I))

print("XY = iZ ?", np.allclose(X @ Y, 1j*Z))
print("YZ = iX ?", np.allclose(Y @ Z, 1j*X))
print("ZX = iY ?", np.allclose(Z @ X, 1j*Y))
print("XZ = -iY ?", np.allclose(X @ Z, -1j*Y))
print("HXH = Z ?", np.allclose(H @ X @ H, Z))
```

12.3 Lab 2: measurement and the Born rule

```
def prob(psi, basis_state):
```

```

    return abs(basis_state.conj() @ psi)**2

def expval(psi, A):
    return np.real(psi.conj() @ A @ psi)

print("Z basis:", prob(plus, ket0), prob(plus, ket1)) # 0.5 0.5
print("X basis:", prob(plus, plus), prob(plus, minus)) # 1.0 0.0
print("<Z> =", expval(plus, Z), " <X> =", expval(plus, X))

```

$|+\rangle$ is a coin flip in Z and perfectly certain in X . Now watch collapse:

```

def collapse(psi, basis_state):
    p = prob(psi, basis_state)
    return (basis_state * (basis_state.conj() @ psi)) / np.sqrt(p)

after = collapse(plus, ket0)
print("state after measuring 0:", after) # exactly |0>
print("measure Z again :", prob(after, ket0)) # 1.0 -> repeatable
print("but measure X now :", prob(after, plus)) # 0.5 -> superposition gone

```

12.4 Lab 3: the Bloch sphere

```

def bloch(psi):
    return np.array([expval(psi, X), expval(psi, Y), expval(psi, Z)])

def from_angles(theta, phi):
    return np.array([np.cos(theta/2),
                    np.exp(1j*phi)*np.sin(theta/2)], dtype=complex)

for th, ph, name in [(0, 0, "|0>"), (np.pi, 0, "|1>"),
                    (np.pi/2, 0, "|+>"), (np.pi/2, np.pi, "|->"),
                    (np.pi/2, np.pi/2, "|+i>")]:
    r = bloch(from_angles(th, ph))
    print(f"{name:5s} r = {r} |r| = {np.linalg.norm(r):.3f}")

# the overlap formula from Section 11
print("direct :", prob(ket0, plus)) # 0.5
print("Bloch :", (1 + bloch(ket0) @ bloch(plus)) / 2) # 0.5

```

12.5 Lab 4: the double slit as a circuit

Sections 2 and 3, made runnable.

```

def two_path(phi, shots=8192):
    qc = QuantumCircuit(1, 1)
    qc.h(0) # split into two paths
    qc.p(phi, 0) # path length difference
    qc.h(0) # recombine at the screen
    qc.measure(0, 0)
    counts = AerSimulator().run(transpile(qc, AerSimulator()),

```

```

shots=shots).result().get_counts()
return counts.get('0', 0) / shots

for phi in [0, np.pi/4, np.pi/2, 3*np.pi/4, np.pi]:
    print(f"phi = {phi:.2f} P(0) = {two_path(phi):.3f}"
          f" theory = {np.cos(phi/2)**2:.3f}")

```

Fringes: $P(0)$ runs from 1 down to 0 as ϕ goes from 0 to π . Now switch on a which-path detector by copying the path into a second qubit:

```

def two_path_with_detector(phi, shots=8192):
    qc = QuantumCircuit(2, 2)
    qc.h(0)
    qc.cx(0, 1) # <-- the detector: records which path
    qc.p(phi, 0)
    qc.h(0)
    qc.measure([0, 1], [0, 1])
    res = AerSimulator().run(transpile(qc, AerSimulator()),
                             shots=shots).result().get_counts()
    return sum(v for k, v in res.items() if k[-1] == '0') / shots

for phi in [0, np.pi/2, np.pi]:
    print(f"phi = {phi:.2f} P(0) = {two_path_with_detector(phi):.3f}")

```

What You Just Saw

With the detector present, $P(0) \approx 0.5$ for every ϕ . **The fringes are gone.** Nobody looked at the second qubit. It was never printed or inspected. Merely *recording* the path was enough. That is decoherence, in one extra line of code, and it is the entire message of Section 3.

12.6 Lab 5: your first entangled state

```

qc = QuantumCircuit(2, 2)
qc.h(0)
qc.cx(0, 1)
print(qc.draw())

sv = Statevector.from_instruction(qc)
print(sv.data) # [0.707, 0, 0, 0.707] -> (|00> + |11>)/sqrt(2)

qc.measure([0, 1], [0, 1])
counts = AerSimulator().run(transpile(qc, AerSimulator()),
                             shots=4096).result().get_counts()
print(counts) # about {'00': 2050, '11': 2046}, never '01' or '10'

```

The two qubits always agree, yet neither alone has a definite value. Unit 2 gives the tools to say exactly how strange that is.

Problems 12.1–12.5 (computational)

1. Sweep ϕ from 0 to 2π in `two_path` and plot $P(0)$ against ϕ , with $\cos^2(\phi/2)$ overlaid.
2. Rerun `two_path(np.pi/2)` with `shots = 100, 1000, 10000`. How does the scatter around 0.5 shrink with shot count?
3. Write `is_unitary(M)` and test it on X , H , T , and on a matrix that is not unitary.
4. Verify numerically that $|0\rangle\langle 0| = (I + Z)/2$.
5. Replace `qc.cx(0,1)` in Lab 4 with `qc.cry(alpha,0,1)` and try $\alpha = 0, \pi/4, \pi/2, \pi$. Watch the fringes fade gradually as the detector improves, and relate this to $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$.

Unit 1 Summary

Formula Sheet

Amplitudes. $p_i = |c_i|^2$; $|A_1 + A_2|^2 = |A_1|^2 + |A_2|^2 + 2\operatorname{Re}(A_1^* A_2)$.

Interference. $\Delta x = \lambda L/d$; $\lambda = h/p$; $\mathcal{D}^2 + \mathcal{V}^2 \leq 1$.

Dirac. $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$; $\sum_i |e_i\rangle\langle e_i| = I$; $(AB)^\dagger = B^\dagger A^\dagger$.

Operators. Hermitian $A = A^\dagger$ (observables, real eigenvalues); unitary $U^\dagger U = I$ (gates, reversible); $U = e^{-iHt/\hbar}$.

Postulates. (1) unit vector, up to global phase. (2) $|\psi\rangle \rightarrow U|\psi\rangle$. (3) $p(m) = |\langle a_m|\psi\rangle|^2$, collapse to $|a_m\rangle$. (4) $\mathcal{H}_1 \otimes \mathcal{H}_2$.

Expectation. $\langle A \rangle = \langle\psi|A|\psi\rangle$; $\Delta A \Delta B \geq \frac{1}{2}|\langle[A, B]\rangle|$.

Paulis. $X^2 = Y^2 = Z^2 = I$; $XY = iZ$ (cyclic); $XZ = -ZX$; $|0\rangle\langle 0| = (I + Z)/2$.

Hadamard. $H = (X + Z)/\sqrt{2}$; $H^2 = I$; $H|0\rangle = |+\rangle$; $HXH = Z$.

Bloch sphere. $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$; $\vec{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle)$, $|\vec{r}| = 1$; $|\langle\psi|\phi\rangle|^2 = (1 + \vec{r}_\psi \cdot \vec{r}_\phi)/2$.

Two qubits. $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$; CNOT: $|a, b\rangle \rightarrow |a, a \oplus b\rangle$; entangled iff $c_{00}c_{11} - c_{01}c_{10} \neq 0$.

No cloning. No unitary copies an arbitrary unknown state. Follows from linearity alone.

Six Mistakes That Cost Marks

1. **Global vs. relative phase.** $e^{i\theta}|\psi\rangle$ is the same state; $|0\rangle + |1\rangle$ and $|0\rangle - |1\rangle$ are not.
2. **Forgetting to conjugate.** $\langle\psi| = \alpha^*\langle 0| + \beta^*\langle 1|$.
3. **Dagger order.** $(AB)^\dagger = B^\dagger A^\dagger$, reversed.
4. **The half-angle.** Orthogonal states are on opposite sides of the Bloch sphere, not at right angles.
5. **Qiskit bit ordering.** Little-endian. Check with a trivial circuit first.
6. **Misquoting uncertainty.** It bounds spread over many runs, not single-shot precision.

Section to Syllabus Map

Syllabus topic	Section
Quantum and classical physics, example	1
Double slit with bullets and waves	2
Double slit with electrons, probability density	3
Dirac notation	4
State of a quantum system	5
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Quantum mathematics (linearity of states and operators)	7
Discrete and continuous base representations	8
Basic postulates of quantum mechanics	9
Matrix representation of state vectors	10
Geometry of probability distributions, Bloch sphere	11
Python implementation of basic quantum examples	12

What Comes Next

Unit 1 described states as vectors. That works when the system is isolated and you know exactly what was prepared. Two situations break it:

- you do not know which state was prepared (a source flipping a coin);
- you hold one half of an entangled pair, and no vector describes your half.

Unit 2 introduces the **density matrix** ρ , which handles both. Every formula here survives: $\langle \psi | A | \psi \rangle$ becomes $\text{tr}(\rho A)$. The Bloch sphere fills in to a solid **ball**, and decoherence becomes a precise statement about off-diagonal entries shrinking.

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3. R. P. Feynman, *Lectures on Physics*, Vol. III, Chapter 1 — the double slit.
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